

functions, it has the potential to someday replace the CPU.

But that is not something Intel will allow so easily. In fact, there have been legal battles that Intel has fought with NVIDIA, AMD and Federal Trade Commission (FTC), where Intel had to settle and pay. (The FTC is there to protect the consumer and to prevent anti-competitive activities by companies.)

Since a GPU can carry out some of the functions of a CPU and, at multiple



Fig. 3: VW I.D. Buzz powered by NVIDIA DRIVE IX technology

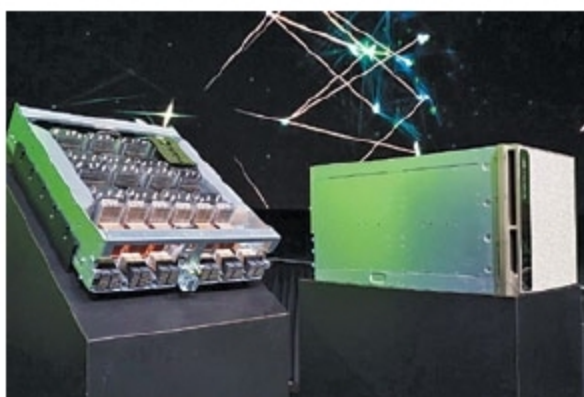


Fig. 4: NVIDIA DGX-2 with computing power to deliver two petaFLOPS



Fig. 5: Zeiss Biogon lenses



Fig. 6: Compilation of Zeiss print ads over the years

times, its power, some interesting uses have come up. This is a key reason the artificial intelligence (AI) revolution has gone into overdrive—creation of super computers with tremendous processing power. And, a GPU is basically a super computer processor.

Smart cities is one such use case. Cameras placed all across the city capture a lot of data, which GPUs analyse. This helps with traffic and parking management, law enforcement and city services, among others.

In healthcare, GPUs can help with medical imaging, analysing genomes and so on.

NVIDIA recognises this as a breakthrough phase. Academicians are pioneering scientific discoveries. They need computing power for their experiments and discoveries. Hence, NVIDIA is convincing these universities to adopt this computing power.

Retail is using AI to optimise the supply chain and user data to increase conversion, provide customised shopping experiences, micro-targeting/pricing and in other areas.

AI touches all aspects of self-driving cars—be it safety, driver alertness, easing traffic congestion or designing new products. NVIDIA is being generous with its technology. It is not only working on its own self-driving technology, but is also working with companies like Uber, Baidu, Toyota, Mercedes-Benz, Audi, Volkswagen and Tesla. Clearly, the company believes that partnering is the way forward.

A big thrust area for NVIDIA moving forwards is super computers, which are driving cloud services. All of these are giving up using CPU servers and are moving to GPU ones, including Amazon, Microsoft, Google and Baidu.

Let me give you an idea of how much more powerful a GPU server is in comparison to a CPU server. One GPU server can replace 160 CPU servers. And when you interconnect multiple GPUs, a speed of two petaFLOPS (for

the interested, that is 10^{15} FLOPS) can be achieved. These computers are needed to solve the most complex AI challenges of the future and, frankly, it is mind boggling where GPUs are taking the power of computing.

Takeaway. Understanding market dynamics and consumers is critical. This is what led to NVIDIA's success over other GPU players. Its strategy to continuously provide upgrades at six-month intervals and keep innovation as its core propelled its growth. It pushed its product designs to maximum technical limits, always providing the best end-user performance.

NVIDIA chose its niche well—gamers—too. This audience was smaller than the main consumer market for PCs, but it was knowledgeable, and that kept NVIDIA at the cutting edge of technology. NVIDIA found the right forums to connect with them, either at conferences or through social media, and used tech influencers well. But perhaps the best is yet to come! The age of super computers and super computer processors has put AI into overdrive, and NVIDIA is at the centre of this revolution that has already begun. Will it be the next Intel?

Zeiss

Zeiss has been the camera of choice for generations. It was the ingredient brand for Nokia phones in the 2000s as well, but that was possible because of Zeiss' beginnings. Carl Zeiss opened his workshop in 1846 and focused on making scientific instruments and lenses. He developed the famous Jena optical glass.

By 1861, Zeiss was considered one of the best in Germany. By the end of WWI, it was the world's largest camera production company. And in 1969, Zeiss Biogon lenses were used in Apollo II mission to the moon by NASA. That is the heritage of Zeiss. Always considered a pioneer and leader in lenses!

It is not that Zeiss has had no competition over the years. It has just been more focused on delivering to consumer needs, on keeping up with the market. Over the years, it built a brand. It developed and released campaigns, showcasing new lenses and the picture quality from Zeiss lenses.

Zeiss also entered into collaborations and licensing agreements with other companies like Hasselblad, Rollei,